

RippleWealth

AI-Driven Investment Risk Intelligence Platform

Sprint 0 Deliverable:

Computer Science Capstone Project CS 691

Prof. Kshitij Sharma

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Group 8:

- Sharvani Kadarla
- Deepali Sawant
- Jai Sheel Bhasker Paladugu
- Ashish Reddy Majjigapu

Team Member Roles & Responsibilities

Project Manager & Software Developer

Sharvani Kadarla

- ▶ Define project vision, roadmap, and milestones
- ▶ Plan sprints and assign tasks efficiently
- ▶ Design overall system architecture and data flow
- ▶ Ensure integration across frontend, backend, and AI components
- ▶ Monitor project progress and resolve blockers

Frontend & Visualization Engineer

Deepali Sawant

- ▶ Build interactive user interface using Streamlit (*or alternatively using HTML, CSS, JavaScript, or other frontend frameworks like ReactJS*)
- ▶ Design dashboards and portfolio visualizations
- ▶ Implement Knowledge Graph visualizations for event impact
- ▶ Ensure usability, clarity, and professional UI/UX
- ▶ Integrate frontend with backend APIs and AI outputs

Team Member Roles & Responsibilities

AI / Machine Learning Engineer

Jai Sheel Bhasker Paladugu

- ▶ Design and implement AI-driven causal reasoning models
- ▶ Model Butterfly Event propagation across industries and companies
- ▶ Construct and manage causal graphs for event impact analysis
- ▶ Use Large Language Models (LLMs) for explainable insights
- ▶ Run scenario simulations and portfolio risk analysis

Backend & Data Engineering Lead

Ashish Reddy Majjigapu

- ▶ Develop RESTful APIs using FastAPI
- ▶ Implement data ingestion pipelines from APIs or mock datasets
- ▶ Handle data preprocessing, validation, and storage
- ▶ Support portfolio risk calculations and stress-testing computations
- ▶ Ensure backend performance, scalability, and reliability

Problem Statement

- ▶ Modern investment platforms are reactive — they alert investors only after stock prices change
- ▶ They do not consider early real-world signals like supply chain disruptions, weather events, labor strikes, or logistics delays
- ▶ Current tools focus mainly on financial indicators and ignore non-financial risk triggers
- ▶ Investors lack clear explanations of why their portfolio risk is increasing
- ▶ As a result, users cannot anticipate risk or make proactive portfolio decisions

Project Description

RippleWealth is an AI-driven investment risk intelligence platform that anticipates market risks by detecting real-world disruptions—such as weather events, supply chain delays, and labor strikes—before they appear in stock prices. Unlike traditional investment tools that react only after price changes, RippleWealth provides early insights to help investors understand potential portfolio impacts.

Using a multi-agent AI architecture and LLM-based causal reasoning, the platform maps how these events propagate across industries and impact companies and portfolios. Through an interactive Knowledge Graph and explainable risk metrics like Value-at-Risk (VaR), RippleWealth enables proactive, transparent, and data-driven portfolio risk management.

Persona: Retail Investor (Young Professional)

Ethan Miller is a 29-year-old software engineer and early-career investor who actively manages his personal investment portfolio. He is comfortable with technology and prefers data-driven tools that help him understand market behavior and portfolio risk. He wants simple visual dashboards, scenario simulations, and clear Value-at-Risk insights to support smarter investment decisions. However, he finds most financial tools either too complex or too fragmented. His primary goal is to use RippleWealth to simulate market stress scenarios, understand downside risk, and make safer long-term investment choices.



Persona: Financial Risk Analyst (Professional User)

Sophia Chen is a 38-year-old financial risk analyst working in a high-pressure, data-intensive environment where accurate risk modeling is critical. She regularly evaluates portfolios, market shocks, and inter-company dependencies using quantitative techniques. She needs advanced tools that support Monte Carlo simulations, causal relationship mapping, and large-scale scenario testing. Her main challenge is that existing tools are time-consuming and lack integrated risk propagation analysis. She uses RippleWealth to run deep simulations, analyze systemic risk paths, and generate faster, more explainable risk reports.



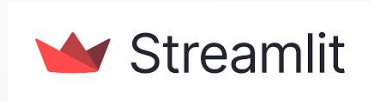
Persona: Business Owner / Wealth Planner

Carlos Ramirez is a 45-year-old small business owner who manages both business and personal investments. He closely follows economic events and sector trends but struggles to understand how global disruptions ripple across industries and affect his holdings. He prefers visual and scenario-based tools rather than purely statistical outputs. His frustration is the lack of clear cause-and-effect visibility in traditional finance platforms. He uses RippleWealth to visualize impact chains, evaluate sector dependencies, and stress-test investment strategies so he can build more resilient financial plans.



Technologies

Core Technologies



Tools and Platforms



Algorithms

Monte Carlo Simulation — Portfolio Risk Modeling

- ▶ Computes portfolio Value-at-Risk (VaR)
- ▶ Simulates thousands of possible market stress scenarios
- ▶ Estimates potential downside loss under uncertainty

Causal Graph Modeling (Graph Theory)

- ▶ Builds event → supply chain → company relationship graphs
- ▶ Identifies direct and indirect impact paths
- ▶ Enables disruption impact tracing across industries

Natural Language Processing (NLP)

- ▶ Extracts “Butterfly Events” from alternative data sources
- ▶ Processes news, weather, and labor reports
- ▶ Converts unstructured text into structured risk signals

Large Language Model (LLM) Reasoning

- ▶ Simulates second- and third-order domino effects
- ▶ Performs causal chain reasoning across sectors
- ▶ Generates explainable impact narratives

Bayesian Risk Propagation

- ▶ Assigns probability weights to impact chains
- ▶ Updates risk exposure dynamically as new signals arrive
- ▶ Supports uncertainty-aware decision modeling

Time-Series Forecasting (ARIMA / LSTM)

- ▶ Predicts short-term volatility trends
- ▶ Supports stress scenario projections
- ▶ Enhances forward risk estimation

Project Schedule and Cadence

SPRINT 0 (Jan 28, 2026 – Feb 17, 2026)

Sprint	Task Name	Priority	Est. Days	Status
Sprint 0	Project Vision Definition	High	2	Done
Sprint 0	Problem Statement	High	3	Done
Sprint 0	Team Roles & Responsibilities	Medium	1	Done
Sprint 0	Personas Definition	Medium	2	Done
Sprint 0	Technology Stack Selection	Medium	2	Done
Sprint 0	GitHub Repository Setup	High	2	Done
Sprint 0	GitHub Wiki Documentation	High	2	Done
Sprint 0	Establish Weekly Meetings	Medium	2	Done
Sprint 0	Work on Deliverable 1	High	5	Done

SPRINT 1 (Feb 18, 2026 – March 3, 2026)

Sprint	Task Name	Priority	Est. Days	Status
Sprint 1	Setup FastAPI Project & Database Mod	High	3	Working on it
Sprint 1	Implement Portfolio & Risk API Endpoint	High	3	Not Started Yet
Sprint 1	Create System Diagrams	High	2	Not Started Yet
Sprint 1	Create acceptance criteria	Medium	1	Not Started Yet
Sprint 1	Create test cases	Medium	1	Not Started Yet
Sprint 1	Create mock up web design page	Medium	1	Not Started Yet
Sprint 1	Create user stories or tasks	High	1	Not Started Yet
Sprint1	Work on Deliverable 2	Medium	2	Not Started Yet

SPRINT 2 (March 4, 2026 – March 24, 2026)

Sprint	Task Name	Priority	Est. Days	Status
Sprint 2	Causal Event Modeling	High	2	Not Started Yet
Sprint 2	Knowledge Graph Visualization Structure	High	2	Not Started Yet
Sprint 2	LLM Reasoning Logic	High	2	Not Started Yet
Sprint 2	Event Impact Simulation	High	2	Not Started Yet
Sprint 2	AI-Backend Integration	High	2	Not Started Yet
Sprint 2	Draft Technical Paper	Medium	2	Not Started Yet
Sprint 2	Develop frontend	Critical	2	Not Started Yet
Sprint 2	Develop backend	Critical	3	Not Started Yet
Sprint 2	Work on Deliverable 3	Medium	4	Not Started Yet

SPRINT 3 (March 25, 2026 – May 12, 2026)

Sprint	Task Name	Priority	Est. Days	Status
Sprint 3	Finish Development of frontend	High	12	Not Started Yet
Sprint 3	Finish Development of backend	High	12	Not Started Yet
Sprint 3	Frontend-Backend Integration	High	4	Not Started Yet
Sprint 3	System Testing	High	4	Not Started Yet
Sprint 3	Finish project demo	Critical	3	Not Started Yet
Sprint 3	Finish technical paper	Critical	7	Not Started Yet
Sprint 3	Work on Deliverable 4	High	7	Not Started Yet

Team Working Agreement

Participation:

- ▶ Attend all scheduled meetings on time
- ▶ Come prepared with progress updates
- ▶ Inform team leader in advance if absent

Meetings:

- ▶ Weekly meeting (Zoom/Google Meet)
- ▶ Team leader schedules and shares meeting links
- ▶ Meeting minutes are recorded after each session
- ▶ Active participation and idea contribution expected

Work Quality & Collaboration:

- ▶ Share work early for feedback
- ▶ Maintain professionalism and accountability
- ▶ Ensure alignment with project goals

Work Division & Ownership:

- ▶ Tasks assigned during sprint planning
- ▶ Equal responsibility distribution
- ▶ Meet agreed deadlines
- ▶ Early communication for blockers
- ▶ Team support during high workload

Communication:

- ▶ Zoom / Google Meet – Sprint meetings
- ▶ WhatsApp Group – Quick updates & urgent communication
- ▶ Google Docs – Collaborative deliverables
- ▶ Project Tools – Track tasks, bugs, attendance, and plans
- ▶ Cameras encouraged for transparency and trust

Retrospective

✓ What Went Well

- Clear communication helped the team align on the RippleWealth vision and problem statement
- Strong collaboration during requirement analysis and persona definition
- Well-defined roles and responsibilities improved ownership and accountability
- Core technologies and tools were identified, giving clear technical direction
- Team recognized early that additional learning is needed for implementation readiness

⚠ What Can Be Improved

- Meeting schedules need better planning to ensure full team availability
- More consistent participation is needed from all members during discussions
- Task breakdown should be more detailed for better effort estimation
- Progress visibility needs to be more structured and regularly updated

Retrospective

Improvement Actions (Added to Sprint 1 Backlog)

Sprint 1 Backlog — Process Improvement Tasks

Action 1 — Structured Mid-Sprint Progress Check-Ins

- Conduct one mandatory mid-sprint review meeting
- Identify blockers early
- Adjust task assignments if needed

Sprint 1 Backlog Task: Mid-Sprint Review Meeting Setup

Action 2 — Better Task Breakdown & Tracking

- Break work into smaller, clearly defined tasks
- Track progress using sprint boards and issues
- Require regular status updates

Sprint 1 Backlog Task: Task Breakdown + Sprint Board Tracking

Retrospective

How We Will Do Things Differently?

- Use Slack as the primary communication platform
- Encourage peer support and knowledge sharing
- Discuss availability and deadlines before each sprint
- Assign clear task ownership to ensure accountability
- Track progress using GitHub, sprint boards and regular updates

Wiki Page Link

<https://github.com/SharvaniKadarla/RippleWealth/wiki>

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Thank You